Gabriel Ricky Noboa

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EDUCATION

University of North Carolina at Charlotte **Expected Graduation:** *May 2028*

BS in Mechanical Engineering, Aerospace Concentration; GPA: 3.5

Organizations: FSAE Aerodynamics Team (EV), Society of Asian Scientists and Engineers (SASE), Rocketry Club

Coursework: Thermodynamics, Mechanics of Solids, Engineering Mechanics, Mechatronics, Manufacturing Processes

RESEARCH EXPERIENCE

OUR SCHOLARS | UNC Charlotte, Charlotte, NC *April 2026 – June 2026*

Undergraduate Researcher, Advisor: Dr. Ankit Agarwal

- Built and tested an adversarial machine learning pipeline in MATLAB to stress-test a CNN (EfficientNet-B0) that classifies CNC tool wear across four failure modes, supporting predictive maintenance in manufacturing
- Designed and benchmarked five attack methods (UAP, DeepFool, targeted, masked spatial, feature collision); fooled the model on 82.5% of images using changes invisible to the human eye
- Found that altering only 34% of an image matched a full-image attack, exposing a key risk in camera-based quality control
- Authored a 19-page research report presented at the OUR Scholars Symposium and accepted for conference publication

PROJECTS

SOLID ROCKET MOTOR NOZZLE | Design and CFD Validation *October 2026*

- Designed a sea-level nozzle for a commercial Cesaroni F29 motor, sizing a 2.93 mm throat and 4.45 expansion ratio from 1D compressible-flow theory at a 400 psi chamber pressure
- Validated certified thrust data to within 1% of rated total impulse in Python, then estimated chamber pressure, mass flow, and throat size where the manufacturer publishes no internal data
- Modeled 15° cone and 80% Rao bell nozzles in SolidWorks and simulated Mach 2.7 flow in SolidWorks Flow Simulation; fixed solver divergence by refining the throat mesh from 14k to 330k cells
- Showed the nozzle delivers 97% of ideal performance (Isp 209–211 s vs. 216 s, discharge coefficient 0.96) and traced the gap to the real motor's 181 s to combustion losses, not nozzle shape
- Found bell and cone within 1.3% of each other at this scale; recommended the simpler cone to cut machining cost

CUSTOM FLIGHT COMPUTER PCB | KiCad *September 2026*

- Took a first-time board from concept to a fabrication-ready layout: schematic capture, footprint management, ERC/DRC checks, and manual trace routing
- Wrote flight-data logging firmware sampling inertial and barometric sensors at 100 Hz for glider flight testing

MOTOR FAULT CLASSIFIER | Python Prototype to C++ Telemetry System *August 2026 – October 2026*

- Built a vibration-sensing test rig (MPU6050 IMU, Arduino, L298N PWM motor control) and trained an FFT-feature classifier reaching 98% accuracy across three motor health states
- Rebuilt the pipeline in multithreaded C++ for real-time acquisition, FFT processing, and SQLite logging, with a Flask dashboard for live and historical telemetry

ACTIVITIES

FSAE CLUB | Aerodynamics Team (EV) *August 2026 – Present*

- Run Star-CCM+ CFD and CAD on aerodynamic components, including front wing airflow analysis

SASE | General Member *August 2026 – Present*

- Mentor underclassmen on career preparation; attend networking and professional development events

SKILLS

Simulation & CFD: ANSYS Fluent (compressible flow, CFD-Post), SolidWorks Flow Simulation, Star-CCM+, nozzle design
CAD: SolidWorks (parts, assemblies, drawings, simulation), CATIA, Creo Parametric, AutoCAD
Programming & Analysis: MATLAB, Simulink, Python (NumPy, SciPy, Matplotlib), C++, SQL, Java, LabVIEW, Git
Machine Learning: CNN image classification (EfficientNet-B0), adversarial robustness testing, predictive maintenance
Embedded & Electronics: KiCad (ERC/DRC, routing), Arduino/ESP32 firmware, IMU and barometric sensor integration
Systems & Tools: Linux (Ubuntu), Flask, Wireshark, TCP/IP, MS Project, Spotfire
Human Languages: English, Spanish, Portuguese (all fluent)